KENZO CHINCHILLA

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EDUCATION

Northeastern University Oakland, CA
Bachelor of Science in Electrical Engineering *Expected 2029*

PROJECTS

DIY SpO2 Pulse Oximeter | *KiCad, Arduino Nano (ATmega328P), MAX30102, OLED, I2C*

- Designed a custom PCB in KiCad, from schematic capture through component placement and trace routing, integrating an Arduino Nano, MAX30102 optical pulse-oximetry sensor, and OLED display
- Created custom footprints for components missing from standard KiCad libraries and generated Gerber files for PCB fabrication
- Interfaced the MAX30102 with the microcontroller over I2C to read red and infrared light measurements used to calculate blood-oxygen saturation (SpO2) and heart rate
- Hand-soldered all components, then tested and debugged the assembled board into a fully working handheld device with real-time readings on the OLED display

FMCW Radar Simulator | *MATLAB* *In Progress*

- Developing a 77 GHz FMCW radar simulator that generates linear chirps (150 MHz bandwidth, 128 chirps per frame) and models echoes from up to five moving targets, including propagation delay, Doppler shift, and Gaussian noise
- Implementing range and Doppler FFTs, 2D CA-CFAR detection, and Kalman filter tracking, evaluating detection and error rates across SNRs from −10 to 20 dB

Pulsed Radar Signal Processor | *Verilog, GTKWave* *In Progress*

- Designing an RTL radar processor that takes in 12-bit samples at 10 MHz, filters noise with a 16-tap fixed-point FIR filter, and detects echoes from up to three targets using threshold and peak detection
- Calculating target range from echo delay and verifying the design with Verilog testbenches in GTKWave, measuring accuracy, latency, and throughput

Drone Detection with Computer Vision | *Python, YOLO*

- Trained a YOLO object-detection model on labeled drone images to locate drones with bounding boxes, testing it on unseen footage

LEADERSHIP & EXPERIENCE

Engineering Club, Northeastern University Oakland 2026 – Present
Project Manager

- Lead the club's engineering projects, coordinating tasks and timelines across team members

MIT Lincoln Laboratory Spring 2024
Participant, LLRISE Radar Program (Accelerated Spring)

- Selected for MIT Lincoln Laboratory's accelerated LLRISE program, focused on radar systems and engineering fundamentals

NASA JPL Invention Challenge, Pioneer Charter School of Science II 2024 – 2026
Team Lead

- One of the main leads on the team's competition prototype for two consecutive years

Pioneer Charter School of Science II 2024 – 2026
Hackathon Organizer

- Organized two school hackathons, managing planning and logistics for student participants

TECHNICAL SKILLS

Languages: Java, Python (basic), C, MATLAB, and Verilog (currently learning)
Hardware: Schematic capture, PCB layout, custom footprints, Gerber export, hand soldering, board assembly and debugging, I2C
Tools & Frameworks: KiCad, Arduino, YOLO, GTKWave